Fundamentals of AI and KR - Module 3

7. Exercises: First part

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Exercise 1

Exercise 1

Which positive test is more informative?

Scenario

A virus is present in 1% of the population. A randomly selected person takes one diagnostic test.

Two tests are available:

	Test A	**Test B**
Sensitivity: $P(+ \mid D)$	0.95	0.90
False positive rate: $P(+ \mid \neg D)$	0.10	0.05

D: the person has the virus.

$+$: the test result is positive.

Question

Which test, when positive, gives a higher probability that the person actually has the virus?

Adapted from Russell and Norvig, AIMA online exercises, Exercise 13.16.

Your turn: identify and calculate

1. Before calculating, which positive result would you consider stronger evidence of infection? Why?
2. Identify the prior probabilities:
$$P(D), \qquad P(\neg D).$$
3. Calculate the probability of a positive result for each test:
$$P(+_A), \qquad P(+_B).$$
4. Calculate and compare the posterior probabilities:
$$P(D \mid +_A), \qquad P(D \mid +_B).$$

Hint

A positive result may be a true positive or a false positive. Account for both possibilities.

Exercise 2

Exercise 2

Which coin was selected?

Scenario

A bag contains three biased coins:

- coin a: probability of heads 0.20;
- coin b: probability of heads 0.60;
- coin c: probability of heads 0.80.

One coin is selected uniformly at random and tossed three times. The same coin is used for all three tosses.

Let $C \in \{a, b, c\}$ identify the selected coin, and let $X_1, X_2, X_3 \in \{H, T\}$ denote the outcomes.

Question

We observe: $X_1 = H$, $X_2 = H$, $X_3 = T$. Which coin is most likely to have been selected?

Adapted from Russell and Norvig, AIMA online exercises, Exercise 14.1.

Your turn: build the model and answer the query

1. Draw a Bayesian network for C, X_1, X_2, X_3.
2. Specify the prior distribution of C and the CPT for each toss.
3. Write the factorization of the joint distribution:

$$P(C, X_1, X_2, X_3).$$

4. Calculate the posterior distribution:

$$P(C \mid X_1 = H, X_2 = H, X_3 = T).$$

5. Are the toss outcomes independent:
   - when the selected coin is known?
   - when the selected coin is unknown?

Exercise 3

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Can we trust the alarm?

Scenario

A nuclear power plant uses a temperature gauge to monitor its core. An alarm sounds when the gauge reports a high temperature. Both the gauge and the alarm can fail. The gauge is more likely to fail when the actual core temperature is high.

We consider five variables:

- T: actual core temperature;
- G: temperature reported by the gauge;
- F_G: the gauge is faulty;
- F_A: the alarm is faulty;
- A: the alarm sounds.

For this exercise, $T, G \in \{normal, high\}$. The other variables are Boolean.

Adapted from Russell and Norvig, AIMA online exercises, Exercise 14.13.

Your turn: build and analyse the network

1. Draw a Bayesian network and motivate each arrow.
2. Write the factorization of the joint distribution.
3. Specify the CPT for G, assuming that the probability of a correct reading is:
   - x when the gauge is working;
   - y when the gauge is faulty.
4. Specify the CPT for A: a working alarm sounds exactly when the gauge reports *high*; a faulty alarm never sounds.
5. Count the free parameters of the network with generic CPTs.
6. Identify the graphical independences involving G and determine its Markov blanket.
7. Identify the graphical independences involving A and determine its Markov blanket.

Exercise 4

Exercise 4

Why is the service unavailable?

Scenario

An online service may become unavailable because of:

- a power problem, P;
- a network problem, N;
- an overload, O.

Each problem can make the service unavailable, but does not necessarily do so: backup mechanisms may prevent the outage. Let $U = \mathit{true}$ mean that the service is unavailable. All four variables are Boolean.

We use a Noisy-OR without leak, with inhibition probabilities:

$$q_P = 0.10, \qquad q_N = 0.20, \qquad q_O = 0.40.$$

Conditional on active causes, their mechanisms fail independently to trigger U.

Application of Noisy-OR, inspired by AIMA Exercise 14.10.

Your turn: combine the causes

1. Draw the Bayesian network.
2. Explain the meaning of $q_N = 0.20$.
3. Calculate:
$$P(U = true \mid P = true, N = true, O = false),$$
$$P(U = true \mid P = true, N = true, O = true).$$
4. What is the probability of unavailability when all three causes are false?
5. Compare the number of free parameters required by a generic CPT and by this Noisy-OR model.
6. Does Noisy-OR require the causes P, N, O to be independent?

Questions?

